

Code Layout, Readability and Reusability

Date	04 October 2026
Team ID	POCKETSMART-TEAM
Project Name	PocketSmart AI – Smart Budget & Recommendation Assistant using Gemini Models
Maximum Marks	3 Marks

Project Structure

```
pocketsmart-ai/
|-- main.py           FastAPI app, routes, session cleanup
|-- auth.py           passwords, JWT, sessions
|-- gemini_utils.py   Gemini calls, fallbacks, links, uploads, history
|-- models.py         Pydantic schemas and validators
|-- templates/
|   |-- _head.html     shared <head> include
|   |-- _navbar_app.html shared navigation include
|   |-- index.html login.html register.html dashboard.html
|   |-- home_planner.html party_planner.html jewelry_planner.html
|   |-- history.html
|-- static/
|   |-- styles.css
|   |-- js/app.js      shared frontend helpers
|   |-- uploads/       uploaded outfit images
|-- tests/test_app.py  API tests (Gemini mocked)
|-- requirements.txt
|-- .env.example
|-- README.md
```

Readability Practices

- Separate routing, authentication, AI logic and schemas into different modules.
- Use descriptive names such as `get_home_recommendations()`, `_generate_with_retry()` and `save_to_history()`.
- Keep HTML in Jinja2 templates and logic in Python and JavaScript.
- Keep the API key and secret in environment variables (`.env.example` documents them).
- Short docstrings at the top of every module and on non-obvious functions.

Reusability

`_generate_with_retry()` is used by all three planners, `_links_for()` builds shopping links for any platform list, and `extract_json_from_response()` is shared. Templates reuse `_head.html` and `_navbar_app.html`, and `static/js/app.js` provides shared helpers (`formatINR`, `apiRequest`, `shoppingLinksHtml`, `fallbackBannerHtml`).

Ownership

Backend Lead – application structure; Frontend Lead – templates, CSS and JavaScript; AI Integration Lead – AI module; Testing & Documentation Lead – schemas and tests.